

Niklas M. Melton

Senior Data Scientist

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Authorized to work in the United States without sponsorship

PROFILE

Senior AI practitioner with a Ph.D. in Computer Science and 9+ years of experience taking systems from research through production. Bridges senior data science, ML/CV software engineering, and applied research, with measurable results across multimodal models, NLP, MLOps, data platforms, financial services, and healthcare.

TECHNICAL SKILLS

Programming: Python, C++, R, MATLAB, SQL, $\LaTeX$

ML/AI: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, transformers, multimodal LLMs, RAG, AI agents, NLP, diffusion models, XGBoost, causal inference

Data & Cloud: pandas, Polars, Spark, Airflow, MySQL, PostgreSQL, SQLite, AWS (S3, EC2, Athena, Glue, SageMaker, EKS, ECS, RDS), Docker

Engineering: Git, CI/CD, GitHub Actions, Jenkins, Poetry, CloudFormation, Prefect, Slurm, model deployment, monitoring, experimentation, technical leadership

PROFESSIONAL EXPERIENCE

Senior Software Engineer - Computer Vision, *Garmin International*

May 2026 - Present

Olathe, Kansas - On-site

- Develop a general-purpose automated annotation system for real-world imagery using multimodal large language models and transformer-based vision models.
- Designed and deployed an authenticated, web-based compute platform for computer-vision workloads, integrating Slurm for job scheduling and resource allocation with Prefect for workflow orchestration on a shared on-premises server used by multiple teams.
- Lead the U.S. computer vision team and coordinate development with a partner engineering team in Taiwan.

Graduate Research Assistant, *Missouri University of Science and Technology*

Aug 2024 - May 2026

Rolla, Missouri - Hybrid

- Researched lifelong learning, adaptive AI, and LLM-enabled workflows across reinforcement, supervised, unsupervised, and deep learning.
- Designed reproducible large-scale experiments on HPC clusters, including training workflows, experiment configuration, evaluation, and results analysis.
- Championed development of a university-wide retrieval-augmented generation system and shaped its retrieval, grounding, and knowledge-integration workflows.
- Supported technology transfer with DARPA L2M and the U.S. Army Night Vision Laboratory; mentored junior researchers whose work contributed to multiple publications.

Senior Blockchain Data Scientist, *OpenBlock Labs*

Apr 2024 - Aug 2024

Remote, United States

- Owned a derivatives incentives product and led a multidisciplinary team operating production systems for 24/7 cryptocurrency markets exceeding \$200M in daily trading volume.
- Automated distribution of more than \$100M in incentives; programs contributed to a 5x increase in client trading volume.
- Built APIs, data pipelines, dashboards, and CI/CD workflows for market data, rewards, experimentation, and production monitoring.

Senior Data Scientist, *Janus Health*

Mar 2022 - Apr 2024

Remote, United States

- Owned *Observe*, a production computer vision product that extracts patient and payer data from screenshots of healthcare documents; increased accurate field-extraction coverage from 20% to 98%.
- Led a two-person applied ML team responsible for the end-to-end extraction pipeline, deployment, A/B testing, data-quality monitoring, and ongoing model performance.
- Applied embedding-based clustering and classification to extracted text and introduced OpenAI API-based pull-request review workflows to improve engineering delivery.

PROFESSIONAL EXPERIENCE (CONTINUED)

Senior Data Scientist, *Tuscan Services*

Aug 2021 - Mar 2022

Kansas City, Missouri - Hybrid

- Served as sole data scientist and data engineer for loan scoring and underwriting, building production APIs, data pipelines, decision dashboards, and model lifecycle controls.
- Enabled 2,000 daily applications and more than \$50M in loan approvals while maintaining validation, auditability, data governance, and federal and state regulatory compliance.
- Replaced rules-based expert systems with ML models that improved repayment rates by 30% and increased profit by 10%.

Lead Artificial Intelligence Engineer, *Guise AI*

May 2017 - Aug 2021

Remote, United States

- Led four engineers developing and deploying computer vision and automated data-collection products, including face and body recognition and an intelligent web crawler.
- Directed MLOps, CI/CD, production support, customer requirements, and contracted annotation teams that built a dataset exceeding 10M labeled human images.

EDUCATION

Ph.D., Computer Science, Missouri University of Science and Technology, 2026

GPA: 3.6/4.0

Dissertation: *Incremental Cluster Validity Indices and Their Role in Interpreting Lifelong Learning Systems*

B.S., Aerospace Engineering, Missouri University of Science and Technology, 2017

OPEN SOURCE & SELECTED RESEARCH

Adaptive Resonance Library (ARTlib), 2024 - Present - Creator and maintainer of a modular Python library implementing 18+ Adaptive Resonance Theory model families. Published in the *Journal of Open Source Software*. GitHub | PyPI | JOSS

Separatix, vertebrae & OverlapIndex - Author of Python packages for pretraining complexity diagnostics, feature-extractor and transfer-learning evaluation, and incremental class/cluster overlap measurement. Separatix | vertebrae | OverlapIndex

Selected publication venues: AAAI, IJCNN, IEEE Access, IEEE CIBCB, and the Journal of Open Source Software.

RECOGNITION

Outstanding Paper Award, IEEE Symposium on Computational Intelligence in Health and Medicine, 2025

Best Presentation Award, International Conference on Soft Computing & Machine Intelligence, 2025